COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	09/02/2026	**DATE SUBMITTED:**	09/02/2026
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SEATWORK 7.2 ALGORITHM ANALYSIS: BEST-CASE AND WORST CASE

1. PROCEDURE OUTPUT

What procedures did you perform?

Instructions: Analyze each algorithm carefully. For every item, determine the following:

1. **Best-Case Time Complexity**
2. **Worst-Case Time Complexity**
3. **Brief explanation** of how you arrived at your answers.

Use asymptotic notation such as **O(1), O(log n), O(n), O(n log n), O(n^2)**, etc.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

None
`for (int i = 0; i < n; i++)`
`{`
`  if (arr[i] == target)`
`  {`
`  return i;`
`  }`
`}`
	BEST CASE: O(n)
	
	WORST CASE: O(n)
		EXPLANATION: target at start or missing.

None
`for (int i = 0; i < n; i++)`
`{`
`  cout << arr[i] << endl;`
`}`
	BEST CASE:O(n)
	
	WORST CASE:O(n)
		EXPLANATION: visits all n items.

None
`for (int i = 0; i < n; i++)`
`{`
	BEST CASE:O(n^2)
	
	WORST CASE:O(n^2)
		EXPLANATION: runs n * n times.

``
    for (int j = 0; j < n; j++)
    {
        cout << i << " " << j << endl;
    }
}
``		

None
``
if (n > 100)
{
    for (int i = 0; i < n; i++)
    {
        cout << arr[i];
    }
}
else
{
    cout << "Small input";
}
``
	BEST CASE: O(1)
	
	WORST CASE:O(1)
		EXPLANATION: constant for n <= 100, linear otherwise.

None
``
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < 10; j++)
    {
        cout << i << j;
    }
}
``
	BEST CASE: O(n)
	
	WORST CASE: O(n)
		EXPLANATION: inner loop constant (10).

None
``
for (int i = 0; i < n; i++)
{
    cout << arr[i];
}
 	
for (int j = 0; j < n; j++)
{
    cout << arr[j] * 2;
}
``
	BEST CASE: O(n)
	
	WORST CASE: O(n)
		EXPLANATION: two loops sum to 2n.

None
``
int x = n;
 	
while (x > 1)
{
    x = x / 2;
}
``
	BEST CASE: : O(log n)
	
	WORST CASE: : O(log n)
		EXPLANATION: input halves each step.

None
`for (int i = 0; i < n; i++)`
`{`
`  for (int j = 0; j < i; j++)`
`  {`
`  cout << i << j;`
`  }`
`}`
	BEST CASE: O(n²)
	
	WORST CASE: O(n²)
		EXPLANATION: inner loop runs n(n-1)/2 times.

None
`for (int i = 0; i < n; i++)`
`{`
`  if (arr[i] == target)`
`  {`
`  cout << "Found";`
`  break;`
`  }`
`}`
	BEST CASE:
	
	WORST CASE:
		EXPLANATION:

None
`for (int i = 0; i < n; i++)`
`{`
`  for (int j = 0; j < n; j++)`
`  {`
`  if (arr[i] == target)`
`  {`
`  break;`
`  }`
`  }`
`}`
	BEST CASE: O(n)
	
	WORST CASE: O(n)
		EXPLANATION: breaks on first check or does n * n.

11. Linear Loop With Logarithmic Inner Work
None
`for (int i = 0; i < n; i++)`
`{`
`  int x = n;`
 	
`  while (x > 1)`
`  {`
`  x = x / 2;`
`  }`
`}`
	BEST CASE: O(n log n)
	
	WORST CASE: O(n log n)
		EXPLANATION: n outer * log n inner.

None
`if (arr[0] == target)`
`{`
`  cout << "Found";`
`}`
	BEST CASE:
	
	WORST CASE:
		EXPLANATION:

``
else
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n;
j++)
        {
            cout << arr[i] +
arr[j];
        }
    }
}
``		

None
``
int max = arr[0];
 	
for (int i = 1; i < n; i++)
{
    if (arr[i] > max)
    {
        max = arr[i];
    }
}
``
	BEST CASE:
	
	WORST CASE:
		EXPLANATION:

None
``
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        for (int k = 0; k < n;
k++)
        {
            cout << i + j + k;
        }
    }
}
``
	BEST CASE:
	
	WORST CASE:
		EXPLANATION:

None ``` for (int i = 0; i < n; i++) ```	**BEST CASE: O(1)** **WORST CASE: O(n log n)**	**EXPLANATION: returns at start or does n * log n.**

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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